1 I

1.1 (1)

The equation of motion is

$$m\begin{pmatrix}\ddot{X}\\ \ddot{Y}\end{pmatrix} = -\nabla U(X,Y) = -m\omega_0^2\begin{pmatrix}X\\ Y\end{pmatrix}.$$

Therefore,

$$\ddot{X} + \omega_0^2 X = 0, \qquad \ddot{Y} + \omega_0^2 Y = 0.$$

1.2 (2)

Applying the initial conditions

$$X(0) = X_0, \quad Y(0) = 0, \quad \dot{X}(0) = 0, \quad \dot{Y}(0) = v_0,$$

we obtain

$$\begin{pmatrix}X(t)\\ Y(t)\end{pmatrix} = \begin{pmatrix}X_0\cos(\omega_0 t)\\ \dfrac{v_0}{\omega_0}\sin(\omega_0 t)\end{pmatrix}.$$

2 II

2.1 (3)

The position vector can be expressed in the rotating basis as

$$\vec{r} = x\vec{e}_x + y\vec{e}_y + z\vec{e}_z.$$

Taking its time derivative in the inertial frame gives

$$\begin{aligned}\vec{V} &= \frac{d\vec{r}}{dt}\\ &= \dot{x}\vec{e}_x + \dot{y}\vec{e}_y + \dot{z}\vec{e}_z + x\dot{\vec{e}}_x + y\dot{\vec{e}}_y + z\dot{\vec{e}}_z\\ &= \vec{v} + \vec{\Omega}\times\vec{r},\end{aligned}$$

where we have used $\dot{\vec{e}}_i = \vec{\Omega}\times\vec{e}_i$.

2.2 (4)

Since $z = 0$ and $\vec{\Omega} = \Omega\vec{e}_z$, we have

$$\vec{V} = (\dot{x} - \Omega y)\vec{e}_x + (\dot{y} + \Omega x)\vec{e}_y.$$

Thus, the Lagrangian is

$$\begin{aligned}L &= T - U\\ &= \frac{m}{2}\left[(\dot{x} - \Omega y)^2 + (\dot{y} + \Omega x)^2 - \omega_0^2(x^2 + y^2)\right]\\ &= \frac{m}{2}\left[\dot{x}^2 + \dot{y}^2 + (\Omega^2 - \omega_0^2)(x^2 + y^2) + 2\Omega(x\dot{y} - y\dot{x})\right].\end{aligned}$$

2.3 (5)

Using the Euler–Lagrange equations,

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = 0, \qquad \frac{d}{dt}\frac{\partial L}{\partial \dot{y}} - \frac{\partial L}{\partial y} = 0,$$

we obtain

$$m\ddot{x} = m(\Omega^2 - \omega_0^2)x + 2m\Omega\dot{y},$$
$$m\ddot{y} = m(\Omega^2 - \omega_0^2)y - 2m\Omega\dot{x}.$$

The terms $m\Omega^2 x$ and $m\Omega^2 y$ correspond to the centrifugal force, while $2m\Omega\dot{y}$ and $-2m\Omega\dot{x}$ correspond to the Coriolis force.

3 III

3.1 (6)

The Lorentz force is

$$\vec{F}_L = q\vec{V} \times \vec{B} = q\vec{v} \times \vec{B} + q(\vec{\Omega} \times \vec{r}) \times \vec{B}.$$

Since $\vec{B} = B\vec{e}_z$, its velocity-dependent part is

$$q\vec{v} \times \vec{B} = -qB\vec{e}_z \times \vec{v}.$$

The Coriolis force is

$$\vec{F}_C = -2m\vec{\Omega} \times \vec{v} = -2m\Omega\vec{e}_z \times \vec{v}.$$

These two terms cancel when

$$qB + 2m\Omega = 0, \qquad \text{so} \qquad \boxed{\Omega = -\frac{qB}{2m}}.$$

3.2 (7)

With this choice of Ω, the velocity-dependent terms cancel. The remaining equation of motion in the rotating frame is

$$\begin{aligned} m\ddot{\vec{r}} &= -m\omega_0^2\vec{r} + m\Omega^2\vec{r} + q(\vec{\Omega} \times \vec{r}) \times \vec{B} \\ &= -m(\omega_0^2 + \Omega^2)\vec{r}, \end{aligned}$$

where $qB\Omega = -2m\Omega^2$ has been used. Hence,

$$\ddot{x} = -(\omega_0^2 + \Omega^2)x, \qquad \ddot{y} = -(\omega_0^2 + \Omega^2)y.$$

Define $\widetilde{\omega} = \sqrt{\omega_0^2 + \Omega^2}$. Since the particle starts at the origin, the initial velocity is the same in the inertial and rotating frames. Therefore,

$$x(0) = y(0) = 0, \qquad \dot{x}(0) = v_0, \quad \dot{y}(0) = 0,$$

and the solution is

$$x(t) = \frac{v_0}{\widetilde{\omega}}\sin(\widetilde{\omega}t), \qquad y(t) = 0.$$

The two coordinate systems are related by

$$X = x\cos(\Omega t) - y\sin(\Omega t),$$
$$Y = x\sin(\Omega t) + y\cos(\Omega t).$$

Thus, the exact solution is

$$X(t) = \frac{v_0}{\widetilde{\omega}} \sin(\widetilde{\omega} t) \cos(\Omega t),$$
$$Y(t) = \frac{v_0}{\widetilde{\omega}} \sin(\widetilde{\omega} t) \sin(\Omega t).$$

Neglecting terms of order B^2 and higher, we obtain

$$\boxed{\begin{aligned} X(t) &= \frac{v_0}{\omega_0} \sin(\omega_0 t) + O(B^2), \\ Y(t) &= -\frac{qBv_0 t}{2m\omega_0} \sin(\omega_0 t) + O(B^2). \end{aligned}}$$

3.3 (8)

For circular motion of radius R, let ω denote the signed orbital angular velocity in the inertial frame. The Lorentz force is

$$q\vec{V} \times \vec{B} = qB\omega\vec{r}.$$

The radial equation of motion is therefore

$$-m\omega^2\vec{r} = -m\omega_0^2\vec{r} + qB\omega\vec{r},$$

which gives

$$\omega^2 + \frac{qB}{m}\omega - \omega_0^2 = 0.$$

For a weak magnetic field, the orbital angular velocity is

$$\omega_B = \omega_{B=0} - \frac{qB}{2m} + O(B^2), \qquad \omega_{B=0} = \pm\omega_0.$$

Since the radius R is the same in both cases,

$$\begin{aligned} E_2 - E_1 &= \frac{mR^2}{2}\left(\omega_B^2 - \omega_{B=0}^2\right) \\ &= -\frac{qB}{2}R^2\omega_{B=0} + O(B^2) \\ &= \boxed{-\frac{qB}{2m}J} + O(B^2), \end{aligned}$$

where $J = mR^2\omega_{B=0}$ is the z component of the orbital angular momentum without the magnetic field. Using the angular momentum with the field instead changes the result only at order B^2.

Alternative Method: Hamiltonian Approach

The Hamiltonian of a charged particle in a magnetic field is

$$H = \frac{1}{2m}(\vec{p} - q\vec{A})^2 + \frac{1}{2}m\omega_0^2(x^2 + y^2),$$

where $\vec{p}$ is the canonical momentum and the vector potential satisfies

$$\nabla \times \vec{A} = \vec{B}, \qquad \partial_x A_y - \partial_y A_x = B.$$

Using the symmetric gauge,

$$\vec{A} = \frac{B}{2}(-y, x, 0),$$

and taking $\vec{p} = (p_x, p_y, 0)$, we obtain

$$\begin{aligned} H &= \frac{1}{2m}\left[\left(p_x + \frac{qB}{2}y\right)^2 + \left(p_y - \frac{qB}{2}x\right)^2\right] + \frac{1}{2}m\omega_0^2(x^2+y^2) \\ &= H_0 + \frac{1}{2m}\left[\frac{q^2B^2}{4}(x^2+y^2) + qB(p_x y - p_y x)\right], \end{aligned}$$

where

$$H_0 = \frac{p_x^2 + p_y^2}{2m} + \frac{1}{2}m\omega_0^2(x^2+y^2).$$

Neglecting terms proportional to B^2, we have

$$\begin{aligned} H &\approx H_0 + \frac{qB}{2m}(p_x y - p_y x) \\ &= H_0 - \frac{qB}{2m}J, \end{aligned}$$

where

$$J = xp_y - yp_x$$

is the canonical angular momentum along the z direction.

Note: Canonical and Mechanical Angular Momentum In the presence of a magnetic field, the canonical momentum differs from the mechanical momentum:

$$\vec{p} = m\vec{v} + q\vec{A}.$$

Therefore, for circular motion of radius R,

$$J = mR^2\omega_B + \frac{qB}{2}R^2.$$

From the equation of motion, we have

$$\omega_B = \omega_{B=0} - \frac{qB}{2m} + O(B^2).$$

Substituting this into the expression for J gives

$$\begin{aligned} J &= mR^2\left(\omega_{B=0} - \frac{qB}{2m}\right) + \frac{qB}{2}R^2 + O(B^2) \\ &= mR^2\omega_{B=0} + O(B^2). \end{aligned}$$

Thus, for circular orbits of the same radius, the canonical angular momentum remains unchanged to first order in B. Consequently, the unperturbed Hamiltonian H_0 also has the same value to this order.

Since the potential energy is unchanged for the same radius R, the difference in total energy is equal to the difference in kinetic energy:

$$\boxed{E_2 - E_1 = -\frac{qB}{2m}J + O(B^2)}.$$

This result can also be interpreted in terms of the interaction between the orbital magnetic moment and the external magnetic field:

$$\Delta E = -\vec{\mu}\cdot\vec{B}, \qquad \vec{\mu} = \frac{q}{2m}\vec{J}.$$